

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of Bachelor of Technology (B.Tech) (EE)

University Examination November 2010

CE215 Object Oriented Programming

Date: 01.12.2010, Wednesday Time: 10:30 a.m to 1:30 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION – I

Q.1(a) Write the statements and mention whether the following are True or False. If false also provide the reason. 05

1. Private members of the same class can directly access by the Class objects using the dot operator.
2. Member functions defined inside a class specifier become inline functions by default.
3. A class should have at least one constructor.
4. Qualification only reduces multiplicity from many to one.
5. A constructor is called whenever an object of the class is used.

Q.1(b) Answer the following questions. 06

1. What is function overloading?
2. Explain Friend function with example.

Q.2 Answer the following questions. 12

1. Write a program to add two complex numbers using operator overloading
2. Write a short note on Operators of C++.

OR

Q.2 Answer the following questions. 12

1. Write a program to swap the private data of a class. Make a function exchange() for swapping. 06
2. What is Inheritance? Explain different types of Inheritance. 04
3. Explain *this* pointer. 02

Q.3 Answer the following questions. 12

1. Explain Inline Function with example.
2. Explain reference variable with example.
3. Explain copy constructor.

OR

Q.3 Answer the following questions. 12

1. Explain New and Delete.
2. Explain structure of C++ Program.
3. Short Note: Benefits of OOP

SECTION – II

Q.4 Do as directed. 11

1. Explain the following commands in MATLAB. 04
(a) whos (b) linspace (c) plot (d) clc
2. Write a computer graphics program to draw a circle in the center of the computer screen having radius 400 units. Also draw two lines passes through its center. Both lines should be at a 90 degree angle to each other. 04
3. What is MATLAB? Draw its schematic diagram with main features. 03

Q.5 Answer the following questions.

12

1. Prepare an object diagram for a graphical document editor that supports grouping, which is a concept used in a variety of graphical editors. Assume that a document is composed of several sheets. Each sheet contains drawing objects, including text, geometrical objects, and groups. A group is simply a set of drawing objects, possibly including other groups. A group must contain at least two drawing objects. A drawing object can be a direct member of at most one group. Geometrical objects include circles, ellipses, rectangles, lines and squares.
2. Draw Event Trace Diagram for ATM scenario.
3. Differentiate:
 - a) Query and Operation
 - b) Association and Aggregation

OR**Q.5 Answer the following questions.**

12

1. Prepare a DFD to find volume and surface area of cylinder.
$$\text{Volume of cylinder} = \pi * r * r * h$$
$$\text{Surface Area of Cylinder} = 2 * \pi * r * h$$
2. What is scenario? What will be the scenario for two party phone call systems?
3. Differentiate:
 - a) Event and State
 - b) Activity and Operation

Q.6 Answer the following questions.

12

1. Can Aggregation be fixed, variable or recursive? Why?
2. What is synergy? Also Define Role Name.
3. Explain: Module and Sheet
4. Define: Computer Graphics. Also explains its components.

OR**Q.6 Answer the following questions.**

12

1. What is the purpose of models in designing? What do you mean by views and stages of a model?
2. Group the following into association, aggregation and generalization:
 - a) A diving philosopher is using a fork
 - b) A polygon is order set of points
 - c) A file is an ordinary file or a directory file.
3. What do you mean by Automatic transition? Also explain edge triggered event.
4. Is dynamic model a collection of state chart diagram? Why? Also separate out the following into active and passive components of DFD: processes, data store object, actor, data flows.
